# STRAT-TRACE

*Source-tagged $^7$Be for process-aware aerosol model evaluation*

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Proposed supervisor: Zhuyun Ye | Part B1 scientific review draft · v9.3

## 1. Excellence

### 1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

**Central question.** Which changes in upper-atmospheric supply and wet removal can surface $^7$Be concentration and deposition records distinguish, and when does improved model agreement conceal compensating errors?

Air-quality models must reproduce not only concentrations, but also the processes determining aerosol persistence and deposition. STRAT-TRACE will use the existing beryllium-7 ($^7$Be) implementation in the Danish Eulerian Hemispheric Model (DEHM) to build a controlled test of process credibility. One modelling system, one radionuclide and a small, fixed experiment set will connect source-resolved budgets to the information contained in real measurements. The intended advance is a **regime-resolved discrimination benchmark**: evidence showing which process explanations are supported, excluded or observationally indistinguishable.

For particle-bound $^7$Be, production, boundary supply, transport, radioactive decay and deposition jointly determine the measured signal [1–3]. Surface air and deposition provide complementary constraints, while production uncertainty and precipitation regime affect their interpretation [1–4]. A source label is an accounting tool, not an additional observation. STRAT-TRACE therefore tests whether model-derived differences survive the actual sampling, uncertainty and independent evaluation, rather than equating sensitivity with identification.

#### State of the art and the specific advance

| Established capability | What STRAT-TRACE will add |
|---|---|
| Joint $^7$Be concentration–deposition evaluation [1,2] and improved production formulations [3]. | A controlled test of when joint observations reject compensating supply–removal explanations, including nuisance-source uncertainty. |
| Source tagging already exists; GEOS-Chem includes a stratospheric-source Be7s tracer [17]. | A verified multi-origin budget linked to discriminatory power at the actual measurement intervals and its dependence on observing regime. |
| Controlled removal contrasts [5] and transport-model comparisons [6]. | A compact within-DEHM intervention set with identical meteorology and blocked evaluation, yielding a protocol reusable without a second model. |

Table 1. Novelty is the integrated, observation-aware test of process explanations; individual tracer and statistical tools are established.

The scientific payoff extends beyond one model's calibration: the benchmark will specify which measurement combinations and regimes are informative for aerosol-process evaluation. The operational relevance is concrete: DEHM contributes to the European CAMS regional air-quality system [18], providing a route for developers to assess the diagnostic method before considering operational changes.

### Measurable objectives, hypotheses and success criteria

| Objective / delivery | Testable hypothesis and evidence |
|---|---|
| **O1 — Source-resolved response** By M9: verified source contributions and budgets. | H1: wet-removal changes produce origin-dependent responses. Compare source fractions and absolute losses across years and solar phases under matched variable/fixed production. Reject differential sensitivity where numerical and sampling uncertainty makes responses indistinguishable. |
| **O2 — Discrimination of explanations** By M15: supply–removal response library. | **H2:** joint air–deposition observations discriminate some scenarios that air-only evaluation cannot. Compare recovery at the same controlled false-discrimination rate; report where adding deposition yields no resolvable gain. |
| **O3 — Transfer and observing value** By M21: held-out tests and discrimination map. | H3: budget-supported explanations retain joint skill across held-out observed years and regimes. Fixed-production controls test source dependence. Reject transfer when gains disappear, uncertainty is excessive, or one observable improves by worsening the other. |

These hypotheses do not presuppose that scavenging is the only deficient process or that descent is too rapid. Upper production, imposed boundary supply, lateral input and measurement representativeness remain alternatives. Success requires a verified response library and a reproducible classification of its observational support; it does not require every regime to identify a unique mechanism. Failure to discriminate will be accompanied by a quantified detection bound and a test of what additional observing information would help.

### Preparation: existing implementation and measurement access

Chham and Ye's existing collaboration has implemented $^{7}$Be in DEHM. Their EGU 2026 abstract documents production mapping, WRF/ERA5 meteorology, aerosol-bound treatment and surface evaluation [7]. The applicant's earlier resolution analysis provides a concrete process question [16]:

| DEHM: 25 km relative to 75 km | Median response |
|---|---|
| Surface $^{7}$Be concentration | +5.9% |
| Bulk deposition: wet + dry | +13.6% |
| Precipitation | −45.9% |

Table 2. Preliminary JJA comparison, 2009–2024, 79 locations [16]. Each response is 100 × (fine/coarse − 1) from paired seasonal means, followed by a spatial median. Locations need not be independent cells. These preparatory configuration contrasts motivate the new experiments; they do not establish observation biases or faster descent.

Greater deposition with lower precipitation motivates testing tracer availability and removal exposure; it does not establish faster transport or a faulty scavenging coefficient. The new work moves from this ambiguity to controlled interventions and observational tests, remaining distinct from the existing implementation/evaluation manuscript.

The applicant confirms 2009–2024 as the currently available experimental deposition baseline; acquisition of additional years is under way [16]. DWD correspondence documents co-located air/deposition sampling and daily precipitation since 2009 [21]. Availability is distinguished from station-level completeness: sample dates, collector changes, gaps and permissions will be inventoried. New records will undergo the same quality control before incorporation. Core delivery does not depend on receipt of additional years.

### Scope protected for a 24-month fellowship

The core uses DEHM alone, with its reduced-chemistry configuration and existing computational access supporting matched experiments over 1986–2024/2025. Variable and fixed production distinguish source variability from atmospheric response across three complete solar cycles. Observational tests use the confirmed 2009–2024 baseline and quality-controlled historical extensions. No new campaign, second radionuclide, GEOS-Chem, HYSPLIT or altered vertical winds are required. Longer runs test process stability, not climate-change attribution.

## 1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices)

### One physical tracer; transparent source and boundary accounting

The reference is the team's reduced-chemistry DEHM–WRF/ERA5 workflow at 75 km, with 29 layers and a top near 100 hPa [7,16]. Transport, mixing, physical decay and wet/dry deposition are retained. The code revision, process operators, aerosol treatment and CRAC:Be formulation [3,9] will be documented and frozen. Observation-facing simulations use realistic time-varying production, with archived solar-modulation and geomagnetic inputs; freezing the formulation does not freeze its forcing series. Layer 1 is the surface; layers increase upward.

Production $P(x,t)$ is partitioned with non-negative masks $w_k$ such that $\Sigma_k w_k = 1$ and $P_k = w_k P$. All labels retain the same transport, physical decay and deposition. Labels record where material enters the simulation, not its subsequent altitude or last tropopause crossing.

| Label | Source partition and interpretation |
|---|---|
| UPPER / MIDDLE / LOWER | Production at $p < 300$ hPa; $300 \le p < 700$ hPa; and $p \ge 700$ hPa, respectively, within atmospheric cells. These are diagnostic bands, not three separate meteorological systems. |
| TOP / LATERAL | Actual external upper and lateral inputs, separately accounted for. Boundary concentrations are converted to the realised transfer; they are not assumed to prescribe constant mass input. |
| INITIAL | Initial inventory propagated through spin-up. Its residual influence is tracked rather than silently attributed to subsequent production. |

**Table 3. Six labels plus an independent total for each production treatment. A short boundary-shift test checks the pre-specified pressure cut-offs.**

UPPER and TOP cannot be called the complete stratospheric contribution: DEHM does not represent the full stratosphere, and pressure alone is not a source-origin observation. Imported material retains a boundary label unless its upstream history is available. Nested-boundary accounting must not double count previously tagged material.

**Production controls.** Companion tracers repeat a time-invariant CRAC:Be field retaining its latitude–altitude structure. TOP/LATERAL concentrations repeat annual climatologies; realised transfers vary with meteorology and are budgeted. If external source variability cannot be controlled, attribution is restricted to in-domain production labels, with boundary contributions separate. Paired source treatments share meteorology and process settings. Interannual differences at matched calendar times diagnose transport and removal together; boundary seasonality is retained. Wet-setting contrasts then test removal. Production-normalized diagnostics are checked against fixed-source results: scalar correction need not remove latitude-, altitude- or lag-dependent source effects [3].

### Numerical verification and process budgets

**Source partition is not mass-balance closure.** The sum of label concentrations, deposition and burdens must reconstruct the independent total. Numerical limiters and boundary updates will be tested explicitly, including reconstruction after a source reweighting. The proposed acceptance target is a relative reconstruction error below 0.1% above a declared absolute floor, and at least ten times smaller than the smallest interpreted response; this is a test criterion, not an achieved result.

For each label k and common control volume, an atom inventory $N_k$ will be checked against:

$$dN_k/dt = P_k + F_{in,k} - F_{out,k} - \lambda N_k - L_{wet,k} - L_{dry,k} + \varepsilon_k.$$

P and F are production and transfer rates, L denotes removal, $\lambda$ is the physical decay constant and $\varepsilon$ is the residual. Fluxes and operator losses will be accumulated at model time steps; pressure-band budgets include moving-boundary/rebinning transfers. Whole-domain closure is mandatory. Uninstrumented local terms will be reported as unresolved rather than inferred from concentration gradients.

Primary outputs are absolute C and D, source fractions $C_k/C_{total}$ and $D_k/D_{total}$ with denominator floors, and integrated wet losses by origin and pressure band. The removal-rate diagnostic $\int L_{wet}\,dt / \int N\,dt$ has inverse-time units, not the meaning of a deposited fraction. Source-reweighted contributions are used only where linear reconstruction has passed; missing boundary and storage terms cannot be absorbed into "transport".

### A fixed experiment matrix over three complete solar cycles

**1986–2024/2025** is the principal simulation window: 39 or 40 calendar years, ending in 2025 where consistent meteorological and production inputs permit, otherwise 2024. It spans complete solar cycles 22–24 (September 1986–December 2019) and the rising and maximum phases of the ongoing, incomplete cycle 25 [22]. The reference and both wet-removal settings share identical meteorology and years. The confirmed observational baseline remains 2009–2024 [16]; historical deposition records are being acquired and enter evaluation only after quality control. Input continuity and usable station intervals are audited separately; multidecadal simulations do not imply complete observational coverage.

| Case | Intervention | Coverage / interpretation |
|---|---|---|
| R + tags | Time-varying CRAC:Be production; reference process operators. | 1986–2024/2025. Reference budgets and direct comparison with available observations. |
| W− / W+ + tags | Multiply the reference aerosol wet-removal rate coefficients by αw = 0.5 / 2; reference αw = 1. | Same full period and production forcing as R. Fixed stress contrasts, not confidence limits or calibrated improvements. |
| S contrasts | Reweight UPPER and TOP contributions separately within each removal setting, after linearity tests. | Matched-tag reconstruction over the full period, separately for each source treatment. Supply changes; transport operators do not. |
| Fixed-source partners | Repeat a common production field; control external source variability as specified above. | All three wet settings, 1986–2024/2025. Interannual atmospheric response under common source forcing. |
| Verification controls | Source linearity, band-boundary and spin-up convergence checks. | Bounded pilots. Spin-up precedes the analysed 1986 start; INITIAL influence is checked. |

**Table 4. Three wet settings over 39/40 years, each with variable- and fixed-production tracers. Their execution and species costs are accounted for in Section 3.1; source reweighting requires verified linearity.**

Wet interventions scale rate coefficients before integration; actual removed mass enters the deposition ledger with positivity maintained. Within each production treatment, prescribed production forcing, source masks, decay, dry deposition, meteorology and transport operators are identical across wet settings. These contrasts test represented removal strength, not a unique scavenging sub-process. The implementation point and test log are pre-submission readiness items.

Supply contrasts retain global production and lateral-input uncertainty as nuisance alternatives. A scalar supply change is not faster transport. Each wet setting and production treatment uses its own verified label fields; reference-field reweighting cannot emulate another removal run or arbitrary time-varying production. Plausible scenarios preserve documented production–boundary dependencies. A supply–removal interaction is a difference of differences in the same output, expressing changing sink exposure.

### Interannual robustness and independent temporal evaluation

Before fitting, classify seasons by rainfall amount, wet-day frequency, persistence and mixing diagnostics. Compare wet/dry years and solar phases across 1986–2024/2025, separating realistic-source and fixed-source responses. Empirical claims are limited to quality-controlled observed intervals; solar-cycle coverage of the simulation is not observational validation of all three cycles. Daily rainfall supports daily/weekly metrics, not sub-hourly intensity. Longer records do not resolve structural confounding or establish climate attribution.

Whole-year, multi-year blocks are assigned to selection and locked evaluation using meteorology, solar phase, coverage and collector-method eras before fitting. All wet settings and source treatments cover both sets; only realistic-source runs are scored against observations. The confirmed 2009–2024 baseline anchors empirical tests. Later-received records enter a pre-registered extension without retuning. Spatial groups account for shared cells; whole-year resampling tests regime dependence. Receptor series and budgets cover the full simulation; detailed 3-D output is episodic.

### Measurements, discrimination tests and interpretation

DWD records will be linked through station history, sample dates, relocations and collection method [21]. The reported 2018 collector transition will be retained as a method stratum: pre-/post-transition results and stable-method subsets will be checked separately, not pooled as a homogeneous record. Only quality-controlled paired samples enter joint tests; air-only stations provide context. Panel size and usable years come from the inventory, not the 79 preliminary model locations.

The observation operator averages air activity over collection windows, flow-weighted when possible, in mBq $m^{-3}$, and integrates deposition over its own windows in Bq $m^{-2}$. Air and deposition windows need not be identical: each is simulated on its actual support. DWD confirms that collector rinsing includes dry material, so these data require a **bulk-deposition** comparison [21]. Older Bq $L^{-1}$ records are converted with matching rain depth (L $m^{-2}$); method codes, uncertainty and decay conventions are retained. Collector–grid surface differences enter the representativeness error model. Nondetections are censored observations; missing values are not zeros.

Realistic, time-varying CRAC:Be response vectors will be evaluated using air only, deposition only and both; fixed-production companions are attribution diagnostics, not direct fits to real observations. Known scenarios generate synthetic measurements with actual gaps, sampling intervals and uncertainty, including cases outside fitted source assumptions. Measurement and representativeness errors, temporal/spatial dependence and model discrepancy are included; covariance matters when rainfall converts deposition units.

Uncertainty-scaled sensitivity vectors and singular values will screen confounded responses. Low-dimensional likelihood profiles [8] will explore source amplitudes and nuisance terms at each simulated wet setting. Wet-response interpolation is not assumed. The discrimination map will report recovery probability for prescribed contrasts and detectable source-amplitude bounds. A planned reporting criterion is ≥80% recovery at ≤5% false discrimination in synthetic tests, with simulation uncertainty; failure to reach it is reported as a bound outside the tested range, not a fabricated threshold.

Parameters, regime definitions and scoring rules will be fixed before unblinding held-out tracer responses. Separate air/deposition errors, joint predictive scores and interval coverage will be compared with R using whole-year blocks and method-stratified checks. A candidate is supported only if its intervention has a verified budget effect and its improvement transfers across held-out years without a compensating deterioration beyond the pre-specified uncertainty tolerance. Otherwise it is rejected or classified as indistinguishable among the tested alternatives.

### Interdisciplinarity, open science and research-content diversity

Environmental radioactivity determines source/decay and collector conventions; transport physics supplies budgets; statistical experiment design tests information content. Their integration is operational: the same source-resolved simulation is passed through the measurement operator and recovery analysis, preventing disconnected modelling and statistical work packages.

A data-management plan by M3 will define provenance, access, licences and retention. The protocol and withheld blocks will be versioned before fitting. Public outputs will include a DOI-archived analysis release, CSV/NetCDF metadata, executable tests and synthetic examples that run without restricted DEHM code or station records. Permitted derived data will accompany manuscripts; private data remain access-controlled. FAIR metadata and open formats support reuse within Europe’s open-science ecosystem [20], without implying formal EOSC onboarding.

Sex and gender are not explanatory variables for the project’s atmospheric inventories, fluxes or numerical interventions; no human exposure, health outcome or participant data are analysed. Sex-disaggregating these environmental outputs would not address the research question. This assessment will be revisited if the research content changes. Geographic and observational representativeness are addressed through explicit coverage and sampling tests. Inclusive mentoring and collaboration are separate measures in Section 1.3. No new radioactive releases or handling are involved.

### 1.3 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

The proposed host is Aarhus University (AU), Department of Environmental Science, Roskilde, with Zhuyun Ye as main supervisor. Her institutional profile documents regional Eulerian modelling and data assimilation [13]; she is a co-author of the CAMS regional production-system description [18]. The applicant reports three previous one-month research visits and an established joint $^{7}$Be–DEHM workflow [16]. This preparation supports rapid progression from implementation to independent experimental design.

| Period / direction | Specific competence and verifiable output |
|---|---|
| M1–M4<br>AU → Chham | Source-tag coding, boundary accounting and numerical verification under Ye's guidance: reviewed code and a passing reconstruction/closure test suite. |
| M4–M12<br>AU ↔ Chham | Controlled process contrasts and focused training in inverse-problem uncertainty: a reviewed synthetic-recovery protocol before real-data fitting; one methods presentation. |
| M8–M18<br>Chham → AU | Radionuclide production, decay, collector conventions and sampling-aware R/Python workflows: two practical sessions with reusable exercises and unit tests. |
| M18–M24<br>Chham-led integration | Lead the held-out assessment, developer demonstration and open release; produce a documented research portfolio and follow-on funding concept. |

Table 5. Training is assessed through executable methods and independent decisions, not attendance alone. A specialist methodological reviewer for uncertainty analysis is a pre-submission hosting item.

Weekly progress meetings and quarterly design reviews will separate technical progress from scientific interpretation. The career-development plan will be agreed in M1 and reviewed every six months. Chham will lead implementation, analysis and documentation; Ye will review DEHM interventions and scientific claims. An independent methodological review of the recovery/scoring protocol is planned before data fitting; its responsible person or training route must be recorded in the hosting plan.

Inclusive practice will include transparent authorship/contribution records, equitable access to presenting and leading tasks, invitations that seek gender-diverse participation in methods sessions, accessible hybrid materials and scheduling compatible with caring responsibilities. Feedback on mentoring and workload will be part of the six-month reviews. These measures concern working conditions and knowledge transfer; they do not substitute for the research-content assessment in Section 1.2.

### 1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

Essaid Chham is a postdoctoral researcher at Universidad Miguel Hernández. The applicant's record reports a PhD in Chemistry (Granada, 2018) and a PhD in Physics (Abdelmalek Essaâdi, 2019), Erasmus Mundus experience and María Zambrano/APOSTD postdoctoral periods [16]. His combination of Monte Carlo modelling, environmental radioactivity and atmospheric analysis matches a project connecting source physics, numerical tracers and observations.

His research on $^{7}$Be prediction [10], sampling-interval effects in CWT/PSCF [11] and Open-AMA software [12] supports measurement-aware evaluation and reproducibility. Existing DEHM, Fortran, R/Python and NetCDF experience, and teaching/supervision reported in his CV [16], provide the implementation base. The fellowship's new competence is defensible process inference and independent model-development leadership, not a repetition of routine data processing.

## 2. Impact

### 2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development

By M24, Chham will own a portable diagnostic protocol, lead its scientific interpretation and demonstrate its use to model developers. The career portfolio will contain reviewed code, two targeted lead-author manuscripts, teaching materials and a follow-on concept for a model-development or environmental-assessment programme. Six-month reviews will assess independent design decisions and user feedback. This provides evidence for academic and atmospheric-modelling roles beyond radionuclide analysis; subsequent multi-model work is a career pathway, not an added fellowship obligation.

### 2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Dissemination will reach atmospheric modellers and environmental-radioactivity researchers; exploitation will prioritise a usable diagnostic method; communication will explain why a model can match measurements for the wrong reasons. Each activity has a defined audience, product and check of usefulness.

| Audience / timing | Product and uptake measure |
|---|---|
| Scientific community<br>M15–M24 | Two manuscripts: multidecadal source/removal budgets with production controls, and observational discrimination/transfer; preprints and open-access routes. At least one conference contribution. Track submissions, DOI releases and feedback, not promised acceptance. |
| AU/DEHM developers<br>M12 and M22 | Demonstrate the benchmark on a documented case and deliver a developer note identifying supported changes, unresolved alternatives and regression tests. Record review comments and reproducibility issues; target one re-run by a user other than the fellow. |
| Monitoring/data specialists<br>M18–M24 | A concise observing-value brief: where paired deposition adds information and when finer sampling would help. Seek DWD and other relevant specialists' feedback; these are intended audiences, not newly committed partners. |
| Wider research users / public<br>M8–M24 | An open worked example, two practical training sessions and an accessible brief on natural radiotracers and model uncertainty. Record attendance, questions and tutorial completion; avoid equating downloads with impact. |

Table 6. Deliverables and engagement targets are project commitments; external adoption, attendance and publication acceptance are not guaranteed.

The analysis layer will have explicit input schemas, units, sample-window metadata, tests and a permitted-use licence. A small synthetic benchmark will reproduce the full workflow without confidential inputs, enabling independent review and later adaptation to other models. Versioned releases will separate host-owned model code from reusable analysis tools. User feedback will be logged and addressed before the final release; the development team is the first target user, not a claimed service-wide deployment.

### 2.3 The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

**Scientific impact.** The project will deliver one verified multi-origin $^{7}$Be framework, three matched wet-removal settings over 1986–2024/2025 with variable/fixed production, and a discrimination map indexed by regime, sampling and uncertainty. Three complete solar cycles broaden the test of source versus atmospheric variability. Observational support remains tied to 2009–2024 and quality-controlled historical extensions. Developers can assess whether concentration agreement conceals deposition errors or source assumptions, and whether explanations transfer across observed years. Reusable observation operators and synthetic tests extend the method beyond DEHM.

**Technological and environmental-assessment pathway.** DEHM is part of the CAMS regional air-quality system [18]. STRAT-TRACE will provide its development community with diagnostic evidence and tests relevant to aerosol transport/removal, then document whether a supported intervention merits further pollutant-specific evaluation. Better-targeted development can reduce uninformative tuning and improve the traceability of model revisions. Developer time, usability and successful independent re-runs will be assessed; forecast improvement must be demonstrated in subsequent operational validation, not assumed from $^{7}$Be alone.

**Societal and European relevance.** More credible aerosol-process evaluation supports the evidence base for cleaner-air and ecosystem assessments, consistent with the EU Zero Pollution Action Plan [19]. The direct fellowship contribution is a transparent diagnostic method and actionable guidance for modellers and monitoring specialists. Potential later benefits depend on validated transfer to the relevant pollutants and assessment systems. FAIR, reusable outputs [20] lower access barriers to this evidence. This is a defined route from methods to environmental services, rather than an unsupported numerical claim about health or economic savings.

## 3. Quality and Efficiency of the Implementation

### 3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

The 24-month critical path links verified labels, matched production/removal experiments over 1986–2024/2025 and held-out evaluation on available observations. Computational access is already available. Section 3.2 documents this access, historical input continuity and technical readiness; M4 verifies labels, source controls and budgets. The five WPs retain the same objectives and effort allocation.

| WP / span / effort | Tasks and measurable output | Milestone |
|---|---|---|
| WP1<br>M1–M4 / 3 PM | Verify labels, source/boundary controls and spin-up; audit 1986–2024/2025 inputs and confirmed/incoming observations; lock temporal blocks. D1: source/measurement protocol and data-management plan. | MS1, M4: numerical checks pass; period, source controls and protocol frozen. |
| WP2<br>M3–M9 / 5 PM | Run tagged R and its fixed-production partner over 1986–2024/2025; assess solar-phase and regime coverage. D2: paired source-treatment datasets and origin budgets. | MS2, M9: reference budgets verified; totals reconstructed. H1 contrasts follow in WP3. |
| WP3<br>M7–M15 / 6 PM | Run W−/W+ and fixed-production partners over the same years; reweight sources and verify budgets. D3: source–removal response library and first manuscript draft. | MS3, M15: H1/H2 assessed; held-out cases retained. |
| WP4<br>M12–M21 / 5 PM | Recovery tests; locked, method-stratified evaluation on 2009–2024 and audited historical extensions; compare source controls. D4: discrimination map, observing brief and second manuscript draft. | MS4, M21: O3 classified, including explicit non-discrimination bounds. |
| WP5<br>M1–M24 / 5 PM | Training, supervision, data stewardship, developer engagement and dissemination. D5: reusable release, methods materials and career portfolio. | MS5, M24: releases complete; two manuscript submissions targeted. |

Table 7. Total researcher effort is 24 person-months (PM). Overlapping calendar spans do not imply multiple full-time researchers. WP2 starts with preparation; production runs follow MS1. External consultation does not control core delivery.

| WP | 1–3 | 4–6 | 7–9 | 10–12 | 13–15 | 16–18 | 19–21 | 22–24 |
|---|---|---|---|---|---|---|---|---|
| WP1 | ● | ● | | | | | | |
| WP2 | ● | ● | ● | | | | | |
| WP3 | | | ● | ● | ● | | | |
| WP4 | | | | ● | ● | ● | ● | |
| WP5 | ● | ● | ● | ● | ● | ● | ● | ● |
| Gate / output | | MS1 | MS2 | | MS3 | | MS4 | MS5 |

Gantt chart. Filled cells indicate activity during all or part of the interval. D1–D5 correspond to MS1–MS5; the data-management plan is due in M3. No secondment or placement lies on the critical path.

#### Execution budget and control of scope

Three wet settings over 39/40 years require 117/120 configuration-years for time-varying production. Concurrent fixed-production partners add species cost; separate execution adds 117/120 diagnostic configuration-years. Pre-1986 spin-up is additional. A representative-month benchmark includes all tags and both source treatments, documenting CPU time, memory and storage. Existing access and reduced chemistry support execution. A separate 20% restart/verification allowance does not absorb companion experiments. All settings share one meteorological archive.

### Delivery risks and scientific decision branches

Non-identification in a particular regime is a scientific outcome with a detection bound, not an unfinished task. A network that supports no empirical discrimination remains a material limitation. The table therefore separates delivery threats from pre-specified interpretive outcomes; likelihood ratings are qualitative residual assessments after mitigation, not measured probabilities.

| Risk / residual likelihood / impact | Trigger, mitigation and retained result |
|---|---|
| Numerical closure or linearity fails<br>Medium / High | Failed ledger or reconstruction tests stop source-fraction claims. Audit boundary, limiter and decay terms; use a verified coarser partition or source-restricted reruns within the costed reserve. Report the numerical limitation if closure cannot be established. |
| Paired records or metadata inadequate<br>Medium / High | Audit station gaps and method changes, including pre-/post-2018 strata. Extra deposition years enter only after receipt and quality control; delays do not block baseline delivery. Missing records are not zeros. Inadequate core coverage requires redesign. |
| Little discrimination across the available network<br>Medium / Medium | Early synthetic tests evaluate the actual design. Report compatible scenarios and contrast bounds, then quantify the value of alternative sampling in the benchmark. Preserve the source/budget results; do not label a unique atmospheric cause. |
| I/O, archive or workflow delays<br>Medium / High | Audit input continuity over 1986–2024/2025 and benchmark all companion species. Use existing access, checkpoints, execution allowance and online sums. Retain matched wet settings and source controls; any unavoidable period revision must update the claimed solar-cycle coverage and locked evaluation. |
| Candidate improvement fails held-out tests<br>Medium / Medium | Separate air/deposition scores and source/discrepancy stress tests expose compensation. Deliver a documented rejected explanation and a regression test, rather than retune on the validation blocks. |

Table 8. Qualitative residual assessments will be reviewed at each milestone. The response library and discrimination bounds remain deliverables when a particular explanation is rejected.

### 3.2 Quality and capacity of the host institutions and participating organisations, including hosting arrangements

AU's Department of Environmental Science develops atmospheric transport, chemistry and deposition models across scales [14,15]. Its DEHM contribution to CAMS [18], together with the joint $^{7}$Be implementation [7], gives the proposed host a direct scientific and user-community fit. The research configuration is not assumed to be identical to operational CAMS settings; transfer will concern tested diagnostics before any operational model change.

Existing computational access and the team's reduced-chemistry DEHM workflow support day-1 execution without a new supercomputing award. Hosting combines historical input preparation, secure storage and the paired production workflow with weekly supervision and quarterly reviews. Chham leads delivery; Ye guides DEHM science. The 1986–2024/2025 archive, full-species benchmark, code/data permissions, storage and methodological review route will be documented before submission. No new field campaign is required.

Hourly receptor series and online pressure-band/regional budgets cover 1986–2024/2025 for both source treatments. Detailed 3-D output is episodic; checkpoints and ledger checks control reruns. Storage optimisation preserves matched periods, source controls and held-out tests. Effort includes metadata, uncertainty analysis and knowledge transfer. Extra models and resolution or transport-operator perturbations remain outside the core.

**PRE-SUBMISSION DOCUMENTATION — remaining items for this review draft.** Record computational access, infrastructure/contact and paired-production performance; document historical meteorology, solar/geophysical forcing, boundary controls and pre-1986 spin-up. Confirm storage, permissions, supervision and uncertainty training; complete the station/method inventory and wet-operator test. Separate the confirmed 2009–2024 observations from records being obtained. Applicant-confirmed access does not replace institutional documentation.

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